

What Is a Logical Theory?

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September 2025, UNAM, Mexico City, Mexico

Abstract

This paper develops a formal framework for the study of logical theories. We begin with the familiar notion of an inference system as a structure consisting of a formal language and a logical consequence relation, and argue that a logical theory must be understood as something more than a specification of a logical system. In particular, I introduce the notion of an *system of applications*, understood as a structure relating mathematical systems, and in particular inference systems, to the topics of applicability. On this basis, we define a theory about logic as a set of claims concerning which logical systems exist and which topics they are correctly applicable to. The framework is illustrated through examples involving classical and non-classical logics, including paraconsistent, intuitionistic, and connexive logics, as well as through positions such as logical monism and logical pluralism.

Introduction

Logic is a discipline concerned with what can be correctly inferred from what. Thus, given a pair $\langle A; B \rangle$ – where A and B may be propositions, sentences, statements, sequents, data sets, or the like – we want to know whether one can be correctly inferred from the other in a certain sense of ‘correctly inferred’.

There are several senses of ‘correctly inferred’ relevant to this discipline, and they often produce diverse answers to the question of whether from A one can correctly infer B . For instance, B may be correctly inferred from A in the sense that: (i) if A is true, then it is impossible for B to be true; (ii) there is a proof from A to B ; (iii) A provides a correct basis to argue for B ; among other possibilities.

Moreover, there are also competing or rival accounts of how *correct inference* functions for the very same (intended) sense of ‘correct inference’. A logical theory (for a given sense of ‘correctly inferred’) postulates a collection of principles about how *correct inference* works. Thus, in the same way that scientific theory states the principles which are true about the domain of a given scientific theory, a logical theory states the principles which are true about how *correct inference* works.

When B can be correctly inferred from A according to a logical theory L , we say that the inference $\langle A; B \rangle$ is *valid* according to L or *L-valid*. In case $\langle A; B \rangle$ is a schematic

inference (an inference with several placeholders for non-logical terms), we say that $\langle A; B \rangle$ is *L-validating* iff all its instances are *L-valid*.

The way in which a logical theory is structured, however, is not quite straightforward. In this paper, I will provide a framework for representing the structure of logical theories.

I will proceed as follows. ...

1 Inference Systems

Conventionally, a logical theory is represented via an *inference system*, which are pairs consisting of a *formal language* and a *relation of logical consequence* – or *consequence relation* for short.

Definition 1.1 (Inference system). A *inference system* is a pair $S = \langle \mathbf{FoS}, \mathbf{Cn}_S \rangle$, where $\mathbf{FoS}$ is a formal language and $\mathbf{Cn}_S$ is a consequence relation. $\boxtimes$

The structure of the formal language $\mathbf{FoS}$ is usually that of a collection of formulae recursively defined from a vocabulary comprising logical and non-logical terms. For the purposes of this paper, a 0th-order – that is, a propositional or sentential – language will suffice constructed in the usual way with the connectives ' $\neg$ ', ' $\vee$ ', ' $\wedge$ ', ' $\rightarrow$ ', ' $\leftrightarrow$ '. The consequence relation $\mathbf{Cn}_S$ has the structure of a collection of schematic inferences, which are the ones marked as validating by the theory being represented via the system S .

The structure of inferences may be conceived in various ways, the basic one being:

$$\langle A_1, \dots, A_n; B \rangle,$$

where $A_1, \dots, A_n, B$ are the sentences, statements, or propositions. In particular, $A_1, \dots, A_n$ are the premises and B is the conclusion of the inference.¹

The order of the premises is relevant in some inference systems. For simplicity's sake, however, we will disregard these cases and define consequence relations as follows.

Definition 1.2 (Consequence relation). $\mathbf{Cn}_S$ is the consequence relation on an inference system S iff $\mathbf{Cn}_S \subseteq 2^{\mathbf{FoS}} \times \mathbf{FoS}$. $\boxtimes$

Now, consider a logical theory S represented by an inference system with the same name $S = \langle \mathbf{FoS}, \mathbf{Cn}_S \rangle$. Let $\langle \mathbf{A}; B \rangle \in \mathbf{Cn}_S$, where $\mathbf{A} \subseteq \mathbf{FoS}$ and $B \in \mathbf{FoS}$. Our logical theory would then state that any inference of the form $\langle \mathbf{A}; B \rangle$ is such that B is correctly inferred from $\mathbf{A}$. Another way to say this is that the inference form $\langle \mathbf{A}; B \rangle$ is validating.

Note that each inference system has its own notion of validity or correct inference. Thus, when referring to the notion of validity of a given system S , we may use the following equivalent expressions: B is S -correctly inferred from $\mathbf{A}$; the inference $\langle \mathbf{A}; B \rangle$ is S -valid; B is an S -consequence of $\mathbf{A}$; B follows from $\mathbf{A}$ S -logically; B S -follows from $\mathbf{A}$; and the like.

¹Regarding sequent calculi inferences, one might define their structure of that as being of the form of normal inferences (in the sense above) or of tuples of the form $\langle \Gamma; B_1, \dots, B_n \rangle$, where $B_1, \dots, B_n$ are formulae and $\langle \Gamma; B_1 \rangle$ is a normal inference.

2 Logical Theses

The framework presented in the last section is more or less the framework used to speak about logical theories. Here, the theses of a logical theory $\mathbf{S}$ correspond to the inference forms in $\mathbf{Cn}_\mathbf{S}$. Consider, for instance, the following inference forms:

$$\langle \mathbf{A} \cup \{A, \neg A\}; B \rangle \quad (\text{ECQ})$$

$$\langle \mathbf{A}; (B \vee \neg B) \rangle \quad (\text{TND})$$

$$\langle \mathbf{A} \cup \{(A \rightarrow B)\}; \neg(A \rightarrow \neg B) \rangle \quad (\text{B1})$$

Each of these inference forms is validating according to some important inference system and its corresponding logical theory and not validating according to some other important inference system and its corresponding logical theory.

The inference form ECQ, *ex contradictione sequitur quodlibet*, is validating in the classical and the intuitionistic logical theories, but not in paraconsistent ones. The form TND, *tertium non datur*, is validating in the classical and in most paraconsistent and connexive logical theories, but not in paracomplete ones, including the intuitionistic theory. Finally, the form B1, *Boethius' first law*, is validating in connexive logical theories, but it is not in most other logical theories.

3 Relations and Operations on Inference Systems

There are several other kinds inference systems, some of which are used for representing a logical theory. In particular, it is possible to establish some relations between inference systems. This is more easily done when they possess the same formal language, which we are going to assume from now on.²

So let $\mathbf{Fo}$ be the formal language of all the inference systems we will discuss in what follows. Hence, it must be the case for all inference systems $\mathbf{S}$ that $\mathbf{Cn}_\mathbf{S} \subseteq \mathbf{Fo}$. The following are distinguished relations among inference systems.

Definition 3.1 (Sublogic). $\mathbf{S}_1 \subseteq \mathbf{S}_2$ iff $\mathbf{Cn}_{\mathbf{S}_1} \subseteq \mathbf{Cn}_{\mathbf{S}_2}$. ✕

Definition 3.2 (Proper sublogic). $\mathbf{S}_1 \subset \mathbf{S}_2$ iff $\mathbf{Cn}_{\mathbf{S}_1} \subset \mathbf{Cn}_{\mathbf{S}_2}$. ✕

Definition 3.3 (Superlogic). $\mathbf{S}_1 \supseteq \mathbf{S}_2$ iff $\mathbf{Cn}_{\mathbf{S}_2} \subseteq \mathbf{Cn}_{\mathbf{S}_1}$. ✕

Definition 3.4 (Proper superlogic). $\mathbf{S}_1 \supset \mathbf{S}_2$ iff $\mathbf{Cn}_{\mathbf{S}_2} \subset \mathbf{Cn}_{\mathbf{S}_1}$. ✕

Moreover, we can do some operations on logical systems sharing a language. In particular, we can extend the operations of set-union and set-intersection into inference systems as follows.

Definition 3.5 (Logic union). $\mathbf{S}_1 \cup \mathbf{S}_2 = \langle \mathbf{Fo}, \mathbf{Cn}_{\mathbf{S}_1} \cup \mathbf{Cn}_{\mathbf{S}_2} \rangle$. ✕

Definition 3.6 (Logic intersection). $\mathbf{S}_1 \cap \mathbf{S}_2 = \langle \mathbf{Fo}, \mathbf{Cn}_{\mathbf{S}_1} \cap \mathbf{Cn}_{\mathbf{S}_2} \rangle$. ✕

²Alternatively, in case we have two inference systems expressed in two different languages, we might establish a translation between those languages.

Note, however, that, in definitions 3.5 and 3.6, are $\mathbf{Cn}_{S_1} \cup \mathbf{Cn}_{S_2}$ and $\mathbf{Cn}_{S_1} \cap \mathbf{Cn}_{S_2}$ are simply the union and intersection, respectively, of the sets of inferences $\mathbf{Cn}_{S_1}$ and $\mathbf{Cn}_{S_2}$. The sets $\mathbf{Cn}_{S_1} \cup \mathbf{Cn}_{S_2}$ and $\mathbf{Cn}_{S_1} \cap \mathbf{Cn}_{S_2}$ do not necessarily contain the inferences that would follow from the inferences in $\mathbf{Cn}_{S_1}$ and $\mathbf{Cn}_{S_2}$.

For instance, it may be the case that, for some formulae A, B, C :

$$\langle A; B \rangle \in \mathbf{Cn}_{S_1}, \quad (1)$$

$$\langle A; \neg B \rangle \in \mathbf{Cn}_{S_2}, \quad (2)$$

$$\langle B, \neg B; C \rangle \in \mathbf{Cn}_{S_3}, \quad (3)$$

but neither:

$$\langle A; C \rangle \notin (\mathbf{Cn}_{S_1} \cup \mathbf{Cn}_{S_2} \cup \mathbf{Cn}_{S_3}) \quad (4)$$

nor:

$$\langle A; C \rangle \notin (\mathbf{Cn}_{S_1} \cap \mathbf{Cn}_{S_2} \cap \mathbf{Cn}_{S_3}). \quad (5)$$

This makes total sense not only because we are dealing with a simple set-theoretical union, but also because it is not specified which kind of logic (or metalogic) should be used in order to obtain 4 or 5 from 1–3.

4 The Meaning of Validity

What does it mean, however, to say that a given inference form is validating or not in a given logical theory? There are several accounts of this. A very popular one [MacFarlane2000] is that the kind of validity that postulates a logical theory is a formal validity: to simplify, the inferences validated by a inference system are so regardless of the subject matter of the inference ³.

³[To be nuanced.]